

Numbers — Extended Summary (theory + examples)

(Continuing in English, same as the previous document — let me know if you'd prefer Spanish)

1. Number sets: why they exist

Numbers were created to answer human needs (counting, measuring, dividing). Each new equation that couldn't be solved forced the creation of a new set:

Set	Symbol	Built to solve...	Example
Natural	$N = \{1, 2, 3, \dots\}$	Counting	"I have 3 sheep"
Integers	Z	$x + a = 0$	$x + 2 = 0 \rightarrow x = -2$
Rationals	$Q = \{a/b, b \neq 0\}$	$bx = a$	$2x = 3 \rightarrow x = 3/2$
Reals	R	Roots like $h^2 = 2$	$\sqrt{2}$ (irrational, not a fraction)
Complex	C	$x^2 + 1 = 0$ (negative discriminant)	$x = \pm i$

Chain: $N \subset Z \subset Q \subset R \subset C$. Each set is a natural extension needed because some equation couldn't be solved in the previous one.

2. Natural numbers and the Principle of Induction

2.1 Well-ordering principle

Every non-empty subset of N has a **smallest element**. This is the structural property that makes induction work — it guarantees there's always a "first domino."

2.2 The Principle of Induction — formal statement

Let $P(n)$ be a property defined for natural numbers. If:

1. **Base case:** $P(1)$ is true.
2. **Induction step:** for any n , if $P(n)$ is true (this assumption is the **induction hypothesis**), then $P(n+1)$ is also true.

→ Then $P(n)$ is true for every natural number n .

Why it works (intuition): it's like a row of dominoes. If the first one falls (base case) and each falling domino guarantees the next one falls (induction step), then all of them fall — even infinitely many.

2.3 Worked example: sum of the first n odd numbers

Claim $P(n)$: $1 + 3 + 5 + \dots + (2n-1) = n^2$

- **Base case (n=1):** $1 = 1^2$ ✓
- **Induction step:** Assume $P(n)$ is true: $1+3+\dots+(2n-1) = n^2$. We want to prove $P(n+1)$:
 $1+3+\dots+(2n-1) + (2n+1) = n^2 + (2n+1) = (n+1)^2$ ✓ Since replacing the first n terms by n^2 (induction hypothesis) and simplifying gives exactly $(n+1)^2$.

Another classic (Gauss): $1+2+3+\dots+n = n(n+1)/2$. Same base case + induction step logic proves it holds for every n . This is the technique behind Gauss's trick of pairing $1+1000$, $2+999$, etc.

2.4 Application: verifying algorithms

Induction isn't just abstract — it's used to *prove an algorithm is correct for any input size n* .
 Example: an algorithm that finds the maximum of a list.

- Base case: for a list of 1 element, the max is trivially that element.
 - Induction step: if the algorithm correctly finds the max of the first n elements, then comparing that result with element $n+1$ correctly gives the max of $n+1$ elements. This reasoning proves correctness for lists of *any* size — you don't need to test millions of cases one by one.
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3. Complex Numbers

3.1 Definition and standard form

A complex number: $z = a + bi$, where:

- $a = \text{real part} \rightarrow \text{Re}(z)$
- $b = \text{imaginary part} \rightarrow \text{Im}(z)$
- $i^2 = -1$ (the imaginary unit)

If $b = 0 \rightarrow z$ is a real number. If $a = 0 \rightarrow z$ is a **pure imaginary** number (e.g. $3i$, $-2i$).

Represented as the point **(a, b)** on the complex plane (x-axis = real, y-axis = imaginary).

Example: $z = 2 + 3i \rightarrow \text{point } (2, 3)$; $z = 3 - 4i \rightarrow \text{point } (3, -4)$.

3.2 Operations in standard form

Addition/subtraction — operate real and imaginary parts separately:

$$\boxed{z_1 = 1+2i, z_2 = 4+3i \rightarrow z_1+z_2 = (1+4) + (2+3)i = 5+5i}$$

Geometrically this is the **parallelogram law**, same as adding vectors.

Multiplication — treat i like a polynomial variable, then apply $i^2 = -1$:

$$\boxed{z_1=1+2i, z_2=4+3i \rightarrow z_1 \cdot z_2 = 4+3i+8i+6i^2 = 4+11i-6 = -2+11i}$$

Conjugate ($\bar{z}$) — flip the sign of the imaginary part: $z=a+bi \rightarrow \bar{z}=a-bi$. Geometrically it's the mirror image across the x-axis.

- Key property: $z \cdot \bar{z}$ is always a non-negative real number — this is what makes division possible.

$$\boxed{z=1+2i \rightarrow z \cdot \bar{z} = (1+2i)(1-2i) = 1^2 + 2^2 = 5 \text{ (real, positive)}}$$

Division — multiply top and bottom by the conjugate of the denominator to eliminate i from it:

$$\boxed{z_1/z_2 = (1+2i)/(3-4i) = [(1+2i)(3+4i)] / [(3-4i)(3+4i)] = (-5+10i)/25 = -1/5 + 2/5 i}$$

3.3 Polar form

Instead of (a,b) , a complex number can be located by:

- **Modulus r** = distance to the origin = $\sqrt{a^2+b^2}$
- **Argument θ** = angle with the positive x-axis (counterclockwise), $\theta \in (-\pi, \pi]$

Written as $z = r_\theta$.

Standard $\rightarrow$ Polar:

- $r = \sqrt{a^2+b^2}$
- $\theta = \arctan(b/a)$ if $a>0$ • $\arctan(b/a)+\pi$ if $a<0, b>0$ • $\arctan(b/a)-\pi$ if $a<0, b<0$

Example: $z = 1+i \rightarrow r=\sqrt{2}, \theta=\arctan(1)=\pi/4 \rightarrow z = \sqrt{2}_\pi(\pi/4)$

Polar $\rightarrow$ Standard: $a = r \cdot \cos \theta, b = r \cdot \sin \theta \rightarrow z = r(\cos \theta + i \sin \theta)$ — this is the **trigonometric form**.

⚠ Polar form is **not unique**: $r_\theta = r_{(\theta+2\pi k)}$ for any integer k (adding full turns doesn't change the point).

Multiplication/division in polar form (much simpler than in standard form!):

- Multiply: $r_{\theta 1} \cdot s_{\theta 2} = (r \cdot s)_{(\theta 1 + \theta 2)} \rightarrow$ multiply moduli, add angles
- Divide: $r_{\theta 1} / s_{\theta 2} = (r/s)_{(\theta 1 - \theta 2)} \rightarrow$ divide moduli, subtract angles

Example: $z_1=2_\pi, z_2=5_\pi(-\pi/2) \rightarrow z_1 \cdot z_2 = 10_\pi(\pi/2)$ (⚠ Addition/subtraction still require converting back to standard form first — the polar shortcut only works for products and quotients.)

3.4 Exponential form (Euler's formula)

Derived from the infinite series of $e^x, \sin(x), \cos(x)$, replacing x with θi :

$$\boxed{\text{Euler's formula: } e^{(\theta i)} = \cos(\theta) + i \cdot \sin(\theta)}$$

This gives a third representation: $z = r \cdot e^{(\theta i)}$, fully equivalent to the polar form ($r_\theta = r \cdot e^{(\theta i)}$).

Example: $z_1=1+i \rightarrow \sqrt{2}_\pi(\pi/4) \rightarrow \sqrt{2} \cdot e^{(\pi/4 i)}$

Operations mirror the polar ones:

- Product: $r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2 \cdot e^{i(\theta_1 + \theta_2)}$
- Division: $(r_1 e^{i\theta_1}) / (r_2 e^{i\theta_2}) = (r_1 / r_2) \cdot e^{i(\theta_1 - \theta_2)}$
- Conjugate: conjugate of $r \cdot e^{i\theta} = r \cdot e^{-i\theta}$
- **Power** (very useful): $(r \cdot e^{i\theta})^n = r^n \cdot e^{in\theta}$ — this is essentially De Moivre's theorem.

3.5 Roots of complex numbers (roots of unity)

Key difference from real numbers: in $\mathbb{C}$, every nonzero number has **exactly n distinct n -th roots** (not just 0, 1, or 2 like in $\mathbb{R}$).

General formula: given $z^n = r \cdot e^{i\theta}$, the n solutions are:

$$\boxed{z = \sqrt[n]{r} \cdot e^{i(\theta/n + 2\pi k/n)}, \text{ for } k = 0, 1, 2, \dots, n-1}$$

Geometric picture: the n roots sit **evenly spaced on a circle of radius $\sqrt[n]{r}$** , forming the vertices of a regular n -sided polygon, each separated by an angle of $2\pi/n$.

Worked example — cube roots of unity ($z^3 = 1$): $1 = 1 \cdot e^{i0} = 1 \cdot e^{i2k\pi}$, $k=0,1,2,3,\dots$

Solving $z^3 = e^{i2k\pi} \rightarrow z = e^{i2k\pi/3}$, for $k=0,1,2$:

- $k=0$: $z = 1$
- $k=1$: $z = e^{i2\pi/3} = -1/2 + (\sqrt{3}/2)i$
- $k=2$: $z = e^{i4\pi/3} = -1/2 - (\sqrt{3}/2)i$

These three points sit on the unit circle, forming an equilateral triangle.

Worked example — cube roots of -8 : $-8 = 8 \cdot e^{i\pi}$ (modulus 8, argument π) $z = \sqrt[3]{8} \cdot e^{i(\pi/3 + 2\pi k/3)} = 2 \cdot e^{i(\pi/3 + 2\pi k/3)}$, $k=0,1,2 \rightarrow$ three points on a circle of radius 2.

4. Quick-reference formulas

Concept	Formula
Induction	$P(1) \text{ true} + [P(n) \rightarrow P(n+1)] \Rightarrow P(n) \text{ true } \forall n$
Modulus	$r = \sqrt{a^2 + b^2}$
Standard $\leftrightarrow$ Polar	$a = r \cdot \cos \theta, b = r \cdot \sin \theta$
Euler	$e^{i\theta} = \cos \theta + i \sin \theta$
Polar product	$r_1 \cdot s_2 = (rs)_{\theta_1 + \theta_2}$
Polar quotient	$r_1 / s_2 = (r/s)_{\theta_1 - \theta_2}$
Power	$(r \cdot e^{i\theta})^n = r^n \cdot e^{in\theta}$
n-th roots	$z = \sqrt[n]{r} \cdot e^{i(\theta/n + 2\pi k/n)}, k=0, \dots, n-1$
$z \cdot \bar{z}$	always real, ≥ 0